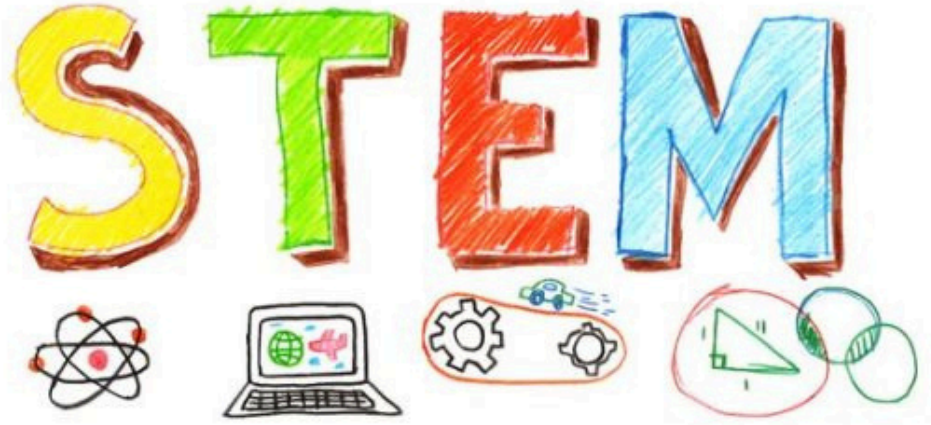

SIMPLE EXPERIMENTS STROLLOUT TO STEM: A SCIENCE ADVENTURE AT HOME

When we think of science, most of us imagine complex laboratories, white coats, and experiments that only specialists can carry out. However, the world of STEM – which stands for Science, Technology, Engineering, and Mathematics – allows for adventure even at home. And what's more, it's fun and informative!

The other day, I made my own "little volcano" in my kitchen. I only used a cup of vinegar, a teaspoon of baking soda, and some red food coloring. As soon as I added the baking soda to the vinegar, the foam that arose filled the kitchen with a "volcanic" explosion. The experiment was so simple and ordinary, yet at the same time impressive, that seeing how fun chemistry concepts can be really got me excited. This simple experiment was a perfect example of how chemistry is all around us and interacts with daily life.

Some other experiments I love to do include the "egg and gravity" experiment. You can put an egg into fresh water, and it will sink to the bottom. However, the outcome will be different if you put the same egg into salty water – it will eventually float.

Experiments with static electricity are just as fun. When you rub a plastic comb or a balloon against your hair, you can see pieces of paper moving. This small experiment is a great way to demonstrate the concepts of electricity and force in a tangible way.

Moreover, designing your own mini experiments means not only learning but also exploring. The best thing about STEM is that you learn from your mistakes. An experiment may not go as expected. The foam might not rise enough, or the egg might not float at a certain level. But in science, there are no mistakes – only learning opportunities.

Asking "Why didn't it work?" nurtures curiosity and the desire to investigate. Sometimes mistakes open the door to unexpected discoveries and guide you toward more creative solutions.

These mini experiments with simple materials at home are a perfect starting point not only for students but for anyone curious. Learning STEM isn't limited to the classroom or the lab; it's about observing, being curious, and experimenting in every corner of life. Starting a small experiment with a glass of water, some vinegar, or baking soda can enhance your scientific thinking skills.

In conclusion, even a small kitchen counter can take you on a big scientific journey. Experiencing STEM in a fun and tangible way increases your curiosity and helps you connect with science. So go ahead, start your own STEM adventure at home with a simple experiment. Who knows – maybe one day these small experiments will pave the way for you to become a great scientist!

by Elif Eda Kesgin